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RESEARCH ARTICLE

A Novel Dense-Swish-CNN With Bi-LSTM Framework for Image Deepfake Detection

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ABSTRACT Artificial intelligence (AI) developments have revolutionized technologies and methodologies, particularly for malicious uses, especially since the advent of generative adversarial networks (GANs) in 2014. GANs can fabricate media streams, seamlessly blending them into different environments. Advanced machine learning techniques can also generate highly realistic deepfake images, which can be exploited for harassment and blackmail. Furthermore, deepfakes can manipulate public sentiment and opinions about individuals or groups. This situation necessitates the development of automated methods to detect deepfakes. In this work, we propose a hybrid deep learning approach for deepfake face detection. The base models include CNN+Bi-LSTM, ResNet34+EfficientNet+Bi-LSTM, and MobileNet+EfficientNet+Bi-LSTM. Initially, images from multiple sources were collected and also generated using various GANs, and the proposed Dense-Swish-CNN with Bi-LSTM was trained. Extract spatial & temporal information from input samples using hybrid pre-trained models. We evaluated our model on challenging datasets such as DFDC, CelebDF, ForgeryNIR, and UADFV. Our experimentation demonstrated the effectiveness of our proposed approach with accuracy 97.76%. Additionally, we assessed the effectiveness of deepfake detection method and validated the explainability of the Dense-Swish-CNN model by performing cross-dataset validation.

INDEX TERMS Biometrics, deepfake, image processing, soft biometrics, swish, Bi-LSTM.

I. INTRODUCTION

In the advancement of deep learning, facial recognition has become an essential biometric feature valued for its simplicity and accuracy. Facial images contain personal information such as gender and emotion and are widely used for personal identification [1], [2]. Recent advances in deep modeling have given rise to new applications ranging from art to entertainment [3]. Nevertheless, similar technology can be used maliciously to create fake photos and videos called deepfakes [1]. Poorly faked content can cause serious harm to people, leading to fake news, business fraud, Trojan horse hoaxes, political fraud, and cyberbullying through synthetic post-red tape [3]. The rapid spread of social media has further increased these problems. Therefore, it is essential to create a more efficient search engine.

Deepfake creations are generally categorized into three types: identity manipulation, expression manipulation, and

attribute transfer [3], [4]. Identity manipulation entails creating fake images of fictional individuals or swapping faces. using FaceSwap or DeepFake. Expression manipulation involves generating facial images with specific expressions or transferring expressions from one person to another [4]. Attribute transfer encompasses altering facial features like age, gender, or hair color [5]. Traditional detection methods relied on manual inspection or forensic analysis, which needed to be faster, more laborious, and often ineffective against advanced deepfake techniques. In recent years, identity manipulation has seen notable advancements.

GANs are used to manipulate face images or videos for malicious purposes, like face morphing attacks and face swapping (deepfakes). State-of-the-art models like PGGAN, StarGAN, and StyleGAN can generate hyper-realistic fake face images using resolutions as high as 1024×1024 . These models have pushed the boundaries of realism in synthetic imagery [5]. Furthermore, recent advancements in expression manipulation techniques have enabled the creation of fake face images with photorealistic effects and

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minimal artifacts. Models such as GANimation and Glow specialize in expression manipulation, offering remarkable results in generating facial expressions with high fidelity [6]. Computer Graphics (CG)-based methods like Face2Face animate facial expressions of a target video from a source actor.

This paper introduces a Swish-based hybrid deep learning algorithm for deepfake detection [3], [4], renowned for its high accuracy and broad applicability. Our method leverages images from diverse sources during model training to enhance its generalization ability. The preprocessing stage involves Gaussian filtering for noise removal, followed by image rescaling before input into the hybrid model. We utilize a binary cross-entropy loss function and the Adam optimizer for the Dense-Swish-CNN model to optimize learning. Employing contourlet transforms, the image is segmented into multiple sub-bands, aiding in detection. Our method extracts pertinent features from the image, including “Local Directional Pattern (LDP), Local Gradient Pattern (LGP), Local Ternary Pattern (LTP), and Local Directional Texture Pattern (LDTP)” [2], [7]. Our hybrid deep learning classifier uses feature selection, fusion, and detection to identify deepfakes, indicating image authenticity or forgery [9].

This paper’s main revolutions and contributions can be summarized as follows:

1. Introduced a novel hybrid deep learning algorithm Dense-Swish-CNN with Bi-LSTM for effective forgery detection.
2. For effective detection of real vs. fake faces the proposed model is trained over 4 diverse datasets.
3. Utilized various GANs to enhance the qualitative representation of image data, improving the model’s ability to detect forgeries (Generalizability).
4. Extensive experiments are conducted to demonstrate the proposed method’s effectiveness, outperforming state-of-the-art methods on various datasets.
5. Performance comparison of CNN models with the Bi-LSTM approach over all the four datasets.

The structure of the paper is as follows: Section II illustrates the deepfake detection literature survey. Section III presents the proposed methodology. The results and discussion are provided in Section IV. Finally, the conclusion of the researched work is given in Section V.

II. LITERATURE REVIEW

The existing literature showcases various approaches researchers have adopted to develop efficient deepfake detection systems. While these approaches differ, their core principles often revolve around identifying inconsistencies and traces of manipulation left by GAN tools during the generation process [10]. Contemporary deepfake techniques extend across diverse modalities, including audio, video, image, or multi-modalities. However, the primary focus of research has been on image/video-based deepfakes, prompting most detection efforts in this direction. As a result, detection models for image/video deepfakes are commonly

categorized into three domains: physical/physiological features, signal level features, and data-driven models [11], [12]. Additionally, some approaches integrate multiple modalities, such as image, audio and video, known as multimodal approaches.

III. FACE MANIPULATION METHODS

Technological advancements have made it increasingly challenging to differentiate between real and fake images. Face manipulation techniques can be categorized into the following types: Entire face synthesis, identity swap, and face reenactment. Face replacement and reenactment can significantly negatively impact society [1]. Entire face synthesis allows for the creation of entirely new facial appearances, often utilized in entertainment and digital avatar applications.

Face identity swap involves transferring entire facial regions from a source to a target image. An early method for this is FaceSwap. To achieve superior results, researchers have proposed methods combining GAN techniques. For instance, Faceswap GAN generates realistic forgery images by leveraging adversarial and perceptual losses [13], [14]. Liu et al. introduced an identity-guided conditional CycleGAN for translating low-resolution face images into high-resolution [14]. They proposed novel multi-scale generator and discriminator architectures, substantially improving the clarity and detail of the generated face images [2].

Face reenactment involves altering attributes like expressions, accessories, or lighting in target face images to match those in source face images. Thies et al. [15] presented Face2Face, a computer graphics-based face reenactment technique. In Face2Face, a 3D model is created for the target faces and source faces. Then, when presented with a pair of target and source faces, the method transfers the expression or attributes from the source face to the target face using these models [15].

A. MANIPULATION DETECTION METHODS

Detecting face manipulation can be viewed as a binary classification task, aiming to discern whether an image is authentic or fake by identifying manipulation artifacts. In recent years, deep learning-based detection methods have demonstrated their superiority in face manipulation detection.

Sabir et al. [16] introduced a recurrent neural network that leverages temporal cues for detecting DeepFake video manipulations. As forgery faces become increasingly realistic, other studies [17], [18] have utilized deep networks to learn discriminative features or identify manipulation traces.

Afchar et al. [19] devised MesoNet, a simple DeepFake detection network that provides interpretable results. Zhang et al. developed a feature set using a bag of words, which offers distinct features for SVM-based binary classification [20]. Chen et al. introduced a multi-stream network design [21], leveraging features of the discrete Fourier transform (DFT).

Kasban et al. [22] introduced the Hilbert–Huang transform for image forgery detection. Ciftci and Demir introduced FakeCatcher, which identifies inauthentic portrait videos using biological signals from facial areas [23]. Huy et al. [24] utilized multi-task learning, developing a convolutional autoencoder network to detect altered videos and simultaneously identify the altered portions for each image input. Additionally, Nguyen et al. [25] proposed a capsule network [11], [32] leveraging features extracted from VGG-19.

IV. PROPOSED METHODOLOGY

A. PROBLEM FORMULATION

The problem formulation for the proposed approach involves predicting the respective class of input images based on their features.

- We define a comprehensive dataset I_{Total} consisting of images $\{I_1, I_2, I_3, \dots, I_n\}$
- This dataset is utilized for training purposes and is segmented into authentic images, I_{real} , and deepfake images I_{fake} .
- The set I_{real} is compiled from p distinct data sources containing x real images. The aggregation of these sources forms the complete set of real images, expressed as:

$$I_{real} = \bigcup_{k=1}^p N_k$$

where N_k represents the set of real images from the k -th data source. If N_k has x_k images, then:

$$I_{real} = \sum_{k=1}^p x_k$$

- Analogous to real images, I_{fake} is aggregated from q separate data sources of deepfake images, with each source containing z images. The deepfake image set is then augmented using many GANs to enhance the qualitative image representation. The complete deepfake image set is represented by:

$$I_{fake} = \bigcup_{j=1}^q z_j$$

- The overall dataset I_{Total} is the union of both real and deepfake image sets, formulated as:

$$I_{Total} = I_{real} \cup I_{fake} = \bigcup_{k=1}^p N_k \cup \bigcup_{j=1}^q z_j$$

Thus, the total number of images is:

$$|I_{Total}| = |I_{real}| + |I_{fake}| = \sum_{k=1}^p N_k + \sum_{j=1}^q z_j$$

Each image in the dataset is assigned a label $Y = \{Y_1, Y_2, Y_3, \dots, Y_m\}$ where m represents the total number of pictures in I_{Total} . The task is framed as a binary classification

challenge, where $y=0$ designates a real image I_{real} and $y=1$ designates a deepfake image I_{fake} . To formalize:

$$\begin{aligned} Y_i &= 0 & \text{If } I_i \in I_{real} \\ Y_i &= 1 & \text{If } I_i \in I_{fake} \end{aligned}$$

for $i=1, 2, \dots, m$.

Algorithm 1 Working of the Proposed Approach

```

Input: I={I1,I2,I3,...,In}
D -> images
M -> models
Output
L={L1,L2,L,...,Ln}
P={P1,P2,P3,...,Pn}
procedure DeepFake_Detection()
  H(t) ← 160 (Height)
  W(t) ← 160 (Width)
  DataGen=ImageDataGenerator(rescale)
  Generator =DataGen.flow_from_directory(D,target_size =
(Ht,Wt),batch_size)
  model ← load_model(M)
  i ← 0
  While i < len(Generator)
    batch ← Generator.next()
    for j in range(len(batch)) do
      Ij ← batch[j][0]
      Lj ← model.predict(Ij[np.newaxis,...])[0][0]
      Pj ← int(Lj > 0.5)
      Display("Likelihood:",Lj)
      Display("PredictedLabel:",Pj)
      Display("Image:",Ij)
    end for
    i ← i + 1
  end while
end procedure

```

Algorithm 1 presents an overview of the proposed architecture for DeepFake detection. The process commences by defining the input image dimensions as 160 pixels in height and width. Subsequently, an ‘ImageDataGenerator’ object, denoted as ‘DataGen’, is instantiated with the requisite parameters for data augmentation. Utilizing this object, images are processed in batches through the ‘flow_from_directory()’ method. This method takes the destination address of stored images and their height and width, resizes them accordingly, and applies necessary data augmentation techniques, yielding a ‘Generator’ object. Next, the pre-trained model is loaded into an object named ‘model.’ Images are fetched from the input directory and fed into the model to predict the likelihood of each image being a DeepFake. The expected likelihood, ranging from 0 to 1, is then rounded to determine the expected label: a value closer to 0 signifies a real image, while a value closer to 1 indicates a DeepFake. A likelihood near 0.5 suggests a random guess. The final output encompasses the predicted

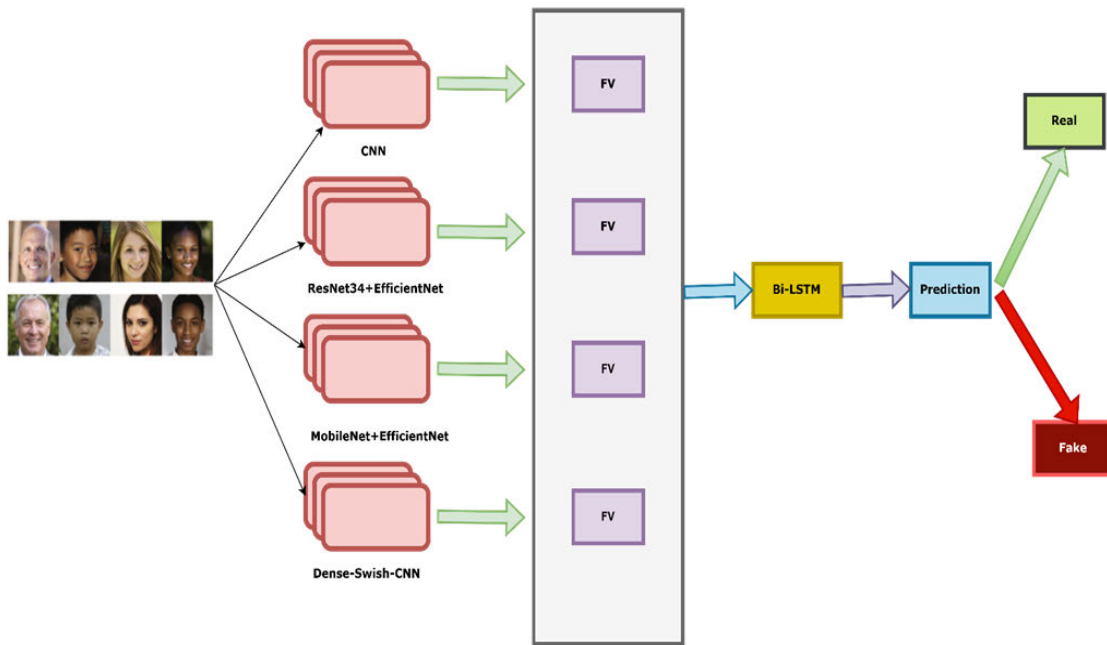


FIGURE 1. The workflow of a proposed deepfake detection system.

likelihood, the predicted label (with '0' representing real and '1' representing DeepFake), and the processed image displayed alongside these results.

B. PROPOSED ARCHITECTURE

This section provides a comprehensive overview of our proposed approach, as depicted in Figure 1. Our method incorporates a convolutional model to handle frame sequences [29], [30], consisting of two key components: i) a CNN unit responsible for extracting deep image features and ii) a Bi-LSTM model for capturing deep spatio-temporal dynamics. The convolutional part leverages several hybrid pre-trained deep learning models [26], specifically CNN+Bi-LSTM, ResNet34+EfficientNet+Bi-LSTM, and MobileNet+EfficientNet+Bi-LSTM. Traditional pre-trained models often fail to capture intricate visual details due to their reliance on a ReLU activation during feature extraction. We propose a unique feature descriptor, Dense-Swish-CNN with Bi-LSTM, to overcome this difficulty by extracting a more robust set of visual characteristics. Feature matrix generated by the convolutional model will be passed to Bi-LSTM by converting it to a 1-D feature vector [33], [34]. Ultimately, the model calculates the probability of an image being a DeepFake or original to generate the final output.

To generate spatial sequence feature vectors capturing visual alterations in input frames (as depicted in Figure 1), we employ a Convolutional Bi-LSTM model [33]. Through end-to-end training, fully connected layers map the high-dimensional Bi-LSTM framework to an output classification score. Our approach integrates two fully connected layers and a dropout layer to address model overfitting [31]. Comprising a CNN phase followed by a Bi-LSTM network,

the Convolutional Bi-LSTM network architecture is detailed in subsequent sections.

C. PRETRAINED MODELS FOR COMPUTATIONS

In this component of the proposed approach, we compute deep key points from input images and then input them into the Bi-LSTM module for deep feature extraction and final classification task, which involves distinguishing between real and Deepfake images. We leverage hybrid deep learning-based pre-trained models, including CNN+Bi-LSTM, ResNet34+EfficientNet+Bi-LSTM, and EfficientNet+MobileNet+Bi-LSTM as base models. The rationale behind using these pre-trained networks in the CNN module of our approach is training on large, publicly available datasets, ensuring robustness and the ability to compute reliable feature sets. Throughout training, the initial layers of these models acquire knowledge about low-level sample key points, while subsequent layers detect and compute task-specific features. Leveraging their extensive expertise and learned image texture information, adapting pre-trained models for deepfake detection reduces training time and improves execution speed, a process commonly referred to as transfer learning. These frameworks adeptly extract features from frames, encompassing facial structure, nose position, eye placement, and lip dimensions [32].

D. DENSE-SWISH-CNN MODEL

The CNN frameworks mentioned earlier only capture the nuanced characteristics of visual samples partially because they utilize the ReLU activation function. ReLU truncates negative input values to zero during keypoint computation, potentially losing essential details. Examined samples may

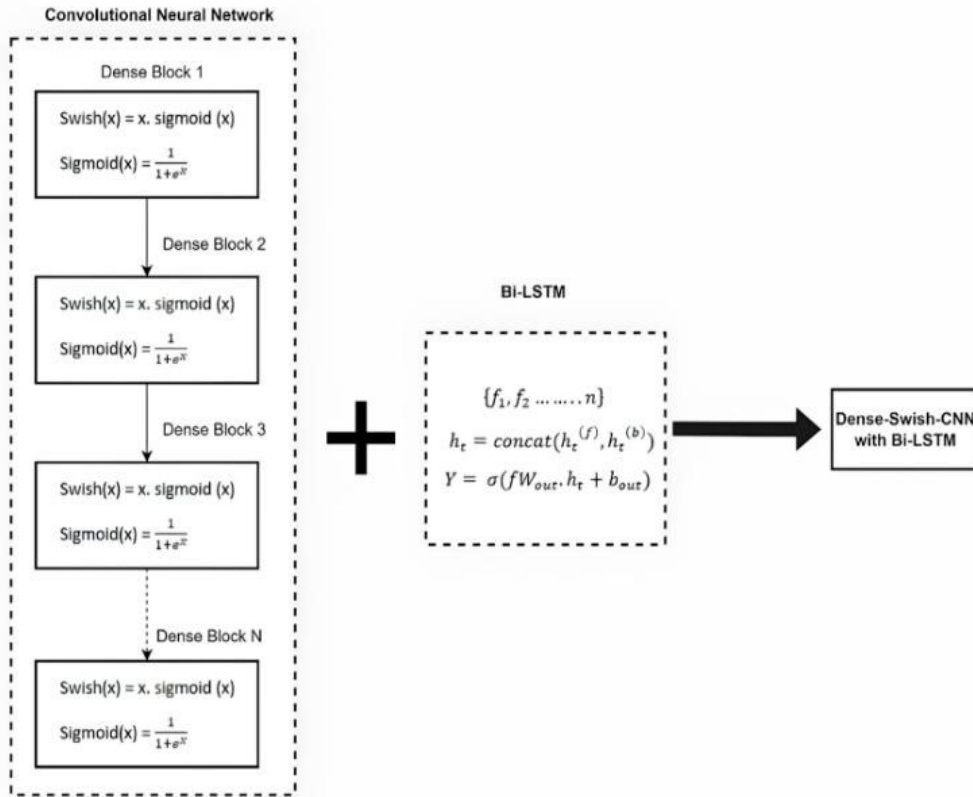


FIGURE 2. Model optimizing process.

lose vital information due to the limitations of these architectures. We introduce a novel framework by optimizing CNN, known as Dense-Swish-CNN to address this limitation and develop a more compelling feature computation strategy for Deepfake detection. This novel framework enhances the standard CNN architecture by incorporating DenseNet principles and replacing the ReLU activation function with the more effective Swish activation, as shown in Figure 2. The extracted spatial features are then fed to Bi-LSTM for temporal feature extraction.

DenseNet will consist of multiple dense blocks, dense blocks in turn consisting of multiple dense layers. Dense block receives input from all the previous convolution layers which improve connectivity by reusing feature maps from previous layers in subsequent ones. Every convolutional layer uses a swish activation function. Within the learning curve, the Swish activation function [27], [28] exhibits smooth, non-monotonic behavior with an unbounded upper limit and a bounded lower threshold. These unique features enable Dense-Swish-CNN to improve its recognition and learning performance while reducing overfitting issues. Compared to the ReLU function, Swish activation is less complex and has demonstrated promising results across various object categorization tasks. Its superiority lies in its ability to permit the flow of negative scores during keypoint computation, unlike ReLU, which inhibits such propagation, thereby discarding essential visual characteristics. By enabling the incorporation

of negative computed scores, the Swish technique facilitates the extraction of a denser and more dependable set of features at the CNN level.

The Dense-Swish-CNN architecture consists of convolutional layers organized into dense blocks, which improve connectivity by reusing feature maps from previous layers in subsequent ones. The model includes convolutional layers with Swish activation functions, followed by batch normalization and pooling operations. After the convolutional layers, the network transitions to fully connected (dense) layers, starting with a 32-unit dense layer, followed by two additional dense layers with 16 units each. These dense layers utilize Swish activation and incorporate dropout for regularization. The generated 1-D vector will be passed to Bi-LSTM. This configuration effectively propagates features throughout the network, resulting in strong performance in detecting image deepfakes.

The proposed architecture processes input images sized 160×160 pixels with a batch size of 45, incorporating data augmentation techniques. A flow diagram is depicted in Figure 3. The initial layers use 2D convolutions with 3×3 and eight filters each, employing Swish activation. Subsequent layers deepen feature extraction with 3×3 convolutions and 16 filters, followed by batch normalization. To manage computational complexity, average pooling layers halve the feature map dimensions. This process repeats with increasing filter sizes and depths, culminating in a 5×5 filter layer

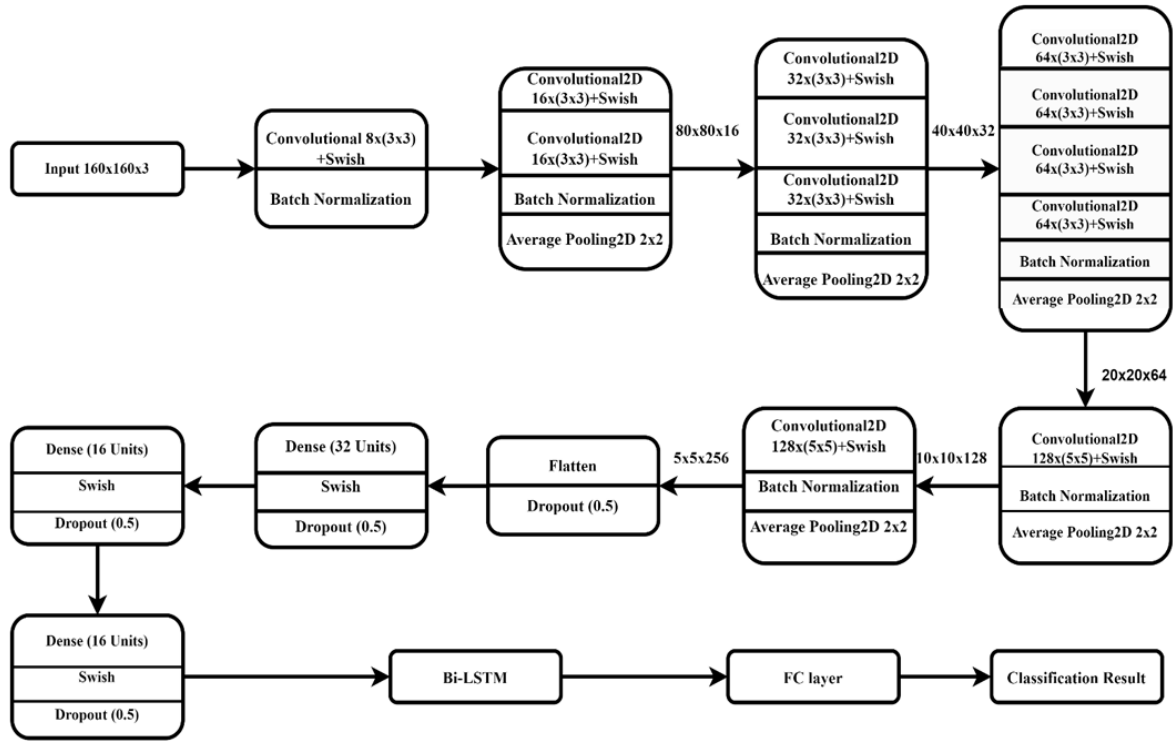


FIGURE 3. Architecture of Dense-Swish-CNN.

with 128 filters, followed by batch normalization and max pooling. The output is then flattened and fed into fully connected layers with Swish activation, followed by dropout layers for regularization. With sigmoid activation, the final layer predicts deepFake authenticity based on the output. Binary cross-entropy loss and the Adam optimizer with a 0.01 learning rate are used during training. The mathematical equations can be represented as follows:

$$Z_1 = \text{Conv2D}(X, W_1) + b_1 A_1 = f(Z_1)$$

$$Z_2 = \text{Conv2D}(A_1, W_2) + b_2$$

$$A_2 = f(Z_2)$$

$$Z_n = \text{Conv2D}(A_{n-1}, W_n) + b_n A_n = f(Z_n)$$

$$P(X) = \sigma(FC(A_n) + b_{out})$$

where, $X \rightarrow$ Image tensor with dimensions

W_i and $b_1 \rightarrow$ Weights and bias of i th layer

$f() \rightarrow$ Swish activation function

$P(X) \rightarrow$ Probability of output

E. BI-LSTM

Following the extraction of a deep feature sequence from an input image's frames, the main goal of the Bi-LSTM model is to classify this sequence as either original or manipulated using a 2-node probability-based network. The Bi-LSTM approach was chosen over other methods, such as optical flow field and 3D-CNN, because of its superior capacity to

analyze temporal behavior; 3D-CNNs are more suited for 3D image analysis, while optical flow only captures short-term motion information. On the other hand, the Bi-LSTM approach effectively represents quality features in time series data, making it a good choice for temporal sequence analysis in video forensic applications. The primary challenge for the Bi-LSTM model lies in processing a sequence iteratively and reliably. To address this, we employed a 1024-wide Bi-LSTM model with a dropout layer of 0.5 to sustain performance and mitigate overfitting. Subsequently, we introduced 512 fully connected layers, leading to a softmax layer with two nodes to assess the probability of a frame sequence being original or deepfake. Initially, images are divided into patches, with each patch treated as a sequence. The convolutional model extracts features from these patches and transfers generated feature matrix to the Bi-LSTM as a 1-D vector. The Bi-LSTM processes these sequences, extracting deeper spatio-temporal features by learning temporal variations in visual appearances over time. This approach enables the Bi-LSTM to classify sequences accurately, enhancing the model's ability to detect subtle temporal changes that may indicate manipulation, such as in DeepFake detection.

V. RESULTS AND DISCUSSION

An extensive examination of the suggested architecture's performance and the outcomes is given in this section. We thoroughly explain the dataset we used in our tests and

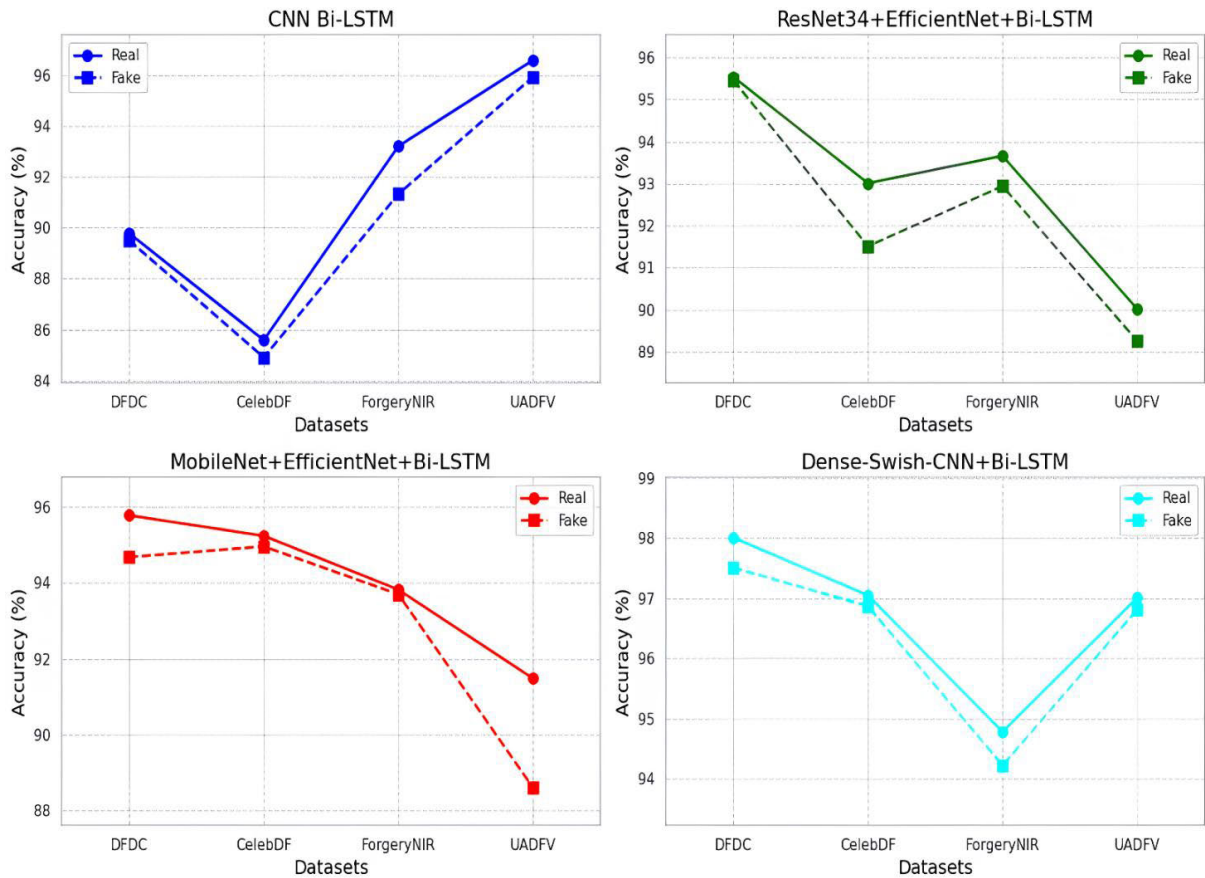


FIGURE 4. Accuracy of proposed models on real and fake data.

discuss the evaluation criteria used to gauge the classification performance of the proposed system.

A. SIMULATION SETUP

The training process utilized Google Colab Pro, typically assigning the Tesla T4 GPU due to restrictions on prolonged GPU usage.

B. DATASET

The dataset comprised authentic images sourced from DFDC, CelebDF, Forgery NIR, and UADFV datasets. Additionally, images were generated from StyleGAN2, CycleGAN, FSGAN, and W-GAN for DeepFake images and given the diverse array of GAN architectures and datasets utilized, images sourced from different origins exhibited resolutions spanning from 1024×1024 to 164×164 , as outlined in Table 1. The image dataset was partitioned, creating train, validation, and test sets. These sets were allocated 70%, 10%, and 20% of the images, respectively, ensuring a balanced representation of images from all sources.

During the training phase, different techniques were used to enhance the model's capacity to identify deepfake images while maintaining its ability to recognize them regardless of their spatial and scale characteristics. These augmentation

TABLE 1. GAN-based dataset.

GAN	Resolution	Count	Type
StyleGAN2	1024x1024	500	Fake
CycleGAN	1024x1024	500	Fake
FSGAN	256x256	500	Fake
W-GAN	164x164	500	Fake

techniques involved flipping the images vertically and horizontally, zooming in by a factor of 0.2, adjusting the shear range by 0.2, shifting the width and height by 0.2, and randomly rotating the images by up to 360 degrees. Integrating these augmentation techniques into the training dataset enabled the model to recognize spatial and scale-invariant features effectively, which is crucial for accurately identifying deepfake images in diverse real-world scenarios. Because most of the faces in the training images were upright, a problem emerged with the Dense-Swish-CNN model's ability to distinguish between deepfakes and genuine images, mainly regarding upright faces. Nevertheless, data augmentation techniques alleviate this risk by offering the

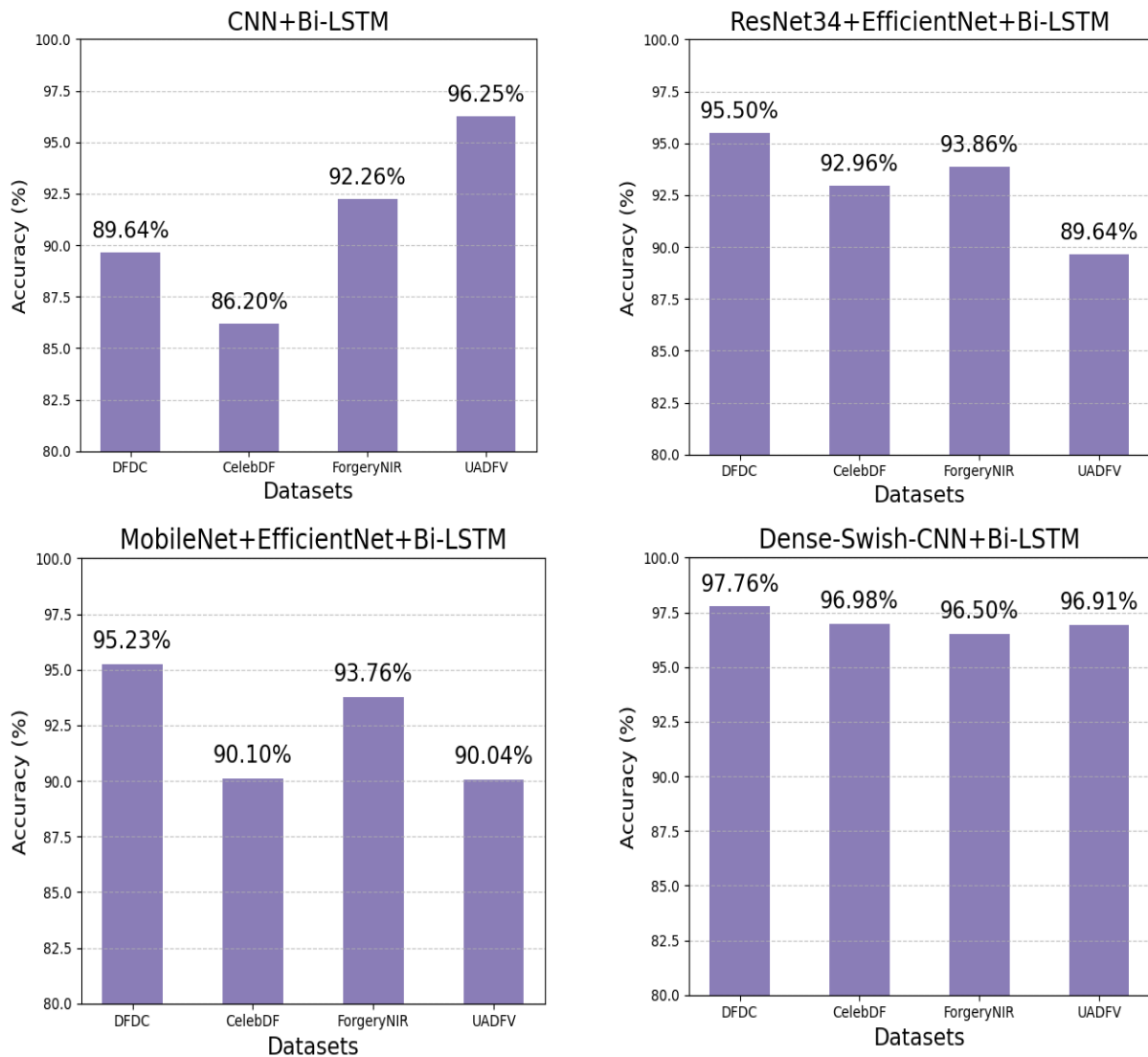


FIGURE 5. Accuracy comparative over datasets.

model a wide range of training samples. Exposing the model to augmented images with different orientations, positions, and transformations can acquire invariant features. Adopting this strategy makes the model more likely to generalize well and perform effectively in various conditions and with different types of images.

Setting the input image size to 160×160 addresses the challenge of detecting low-resolution and low-quality Deep-Fake images. Such images often provide limited information for the model to analyze, making detection more challenging. Additionally, shared social media platforms frequently downscale high-resolution images to reduce transmission and storage costs, further complicating the detection task. Therefore, opting for a CNN network with an input size of $(160, 160, 3)$ aims to ensure the model's practicality in real-world scenarios while minimizing computational costs. During training, we employ the Adam optimizer with a learning rate of 0.01 and training proceeds over 100 epochs.

To cope with hardware and time constraints on platforms like Google Colab, we record training accuracy, precision, F1-score, and recall during the training phase. Furthermore, the batch size is set to 45, significantly stabilizing the training process.

C. EVALUATION OF PRE-TRAINED AND DENSE-SWISH-CNN WITH BI-LSTM MODELS

To illustrate the resilience of our suggested method, we provide the classification outcomes for Deepfakes on each of the four datasets in this section. The proposed model was evaluated using several CNN-based pre-trained approaches, including the CNN Bi-LSTM, ResNet34+EfficientNet+Bi-LSTM, and MobileNet+EfficientNet+Bi-LSTM. We assessed the Deepfake detection performance of these CNN frameworks using images from the employed datasets. Standard evaluation metrics commonly used in image analysis were utilized. Initially, we discuss the results

TABLE 2. Bi-LSTM-based model performance over DFDC.

Model	Accuracy %	Precision %	Recall %	F1-score %
CNN Bi-LSTM	89.64	88.12	87.74	87.63
ResNet34+EfficientNet+Bi-LSTM	95.50	94.78	86.82	94.40
MobileNet+EfficientNet+Bi-LSTM	95.23	96.2	88.76	96.97
Dense-Swish-CNN+Bi-LSTM	97.76	97.5	96.26	96.79

TABLE 3. Bi-LSTM-based model performance over CELEB-DF.

Model	Accuracy %	Precision %	Recall %	F1-score %
CNN Bi-LSTM	86.2	84.32	82.74	82.5
ResNet34+EfficientNet+Bi-LSTM	92.96	91.64	86.5	88.75
MobileNet+EfficientNet+Bi-LSTM	90.10	91.7	92.22	92.10
Dense-Swish-CNN+Bi-LSTM	96.98	96.50	96.38	95.78

TABLE 4. Bi-LSTM-based model performance over ForgeryNIR.

Model	Accuracy %	Precision %	Recall %	F1-score %
CNN Bi-LSTM	92.26	93.5	93.78	92.24
ResNet34+EfficientNet+Bi-LSTM	93.86	92.01	92.05	92.01
MobileNet+EfficientNet+Bi-LSTM	93.76	92.11	92.98	92.98
Dense-Swish-CNN+Bi-LSTM	96.5	94.24	93.35	93.79

TABLE 5. Bi-LSTM-based model performance over UADFV.

Model	Accuracy %	Precision %	Recall %	F1-score %
CNN Bi-LSTM	96.25	95.31	95.35	95.79
ResNet34+EfficientNet+Bi-LSTM	89.64	88.12	87.72	87.98
MobileNet+EfficientNet+Bi-LSTM	90.04	87.99	86.82	87.40
Dense-Swish-CNN+Bi-LSTM	96.91	95.61	95.59	95.60

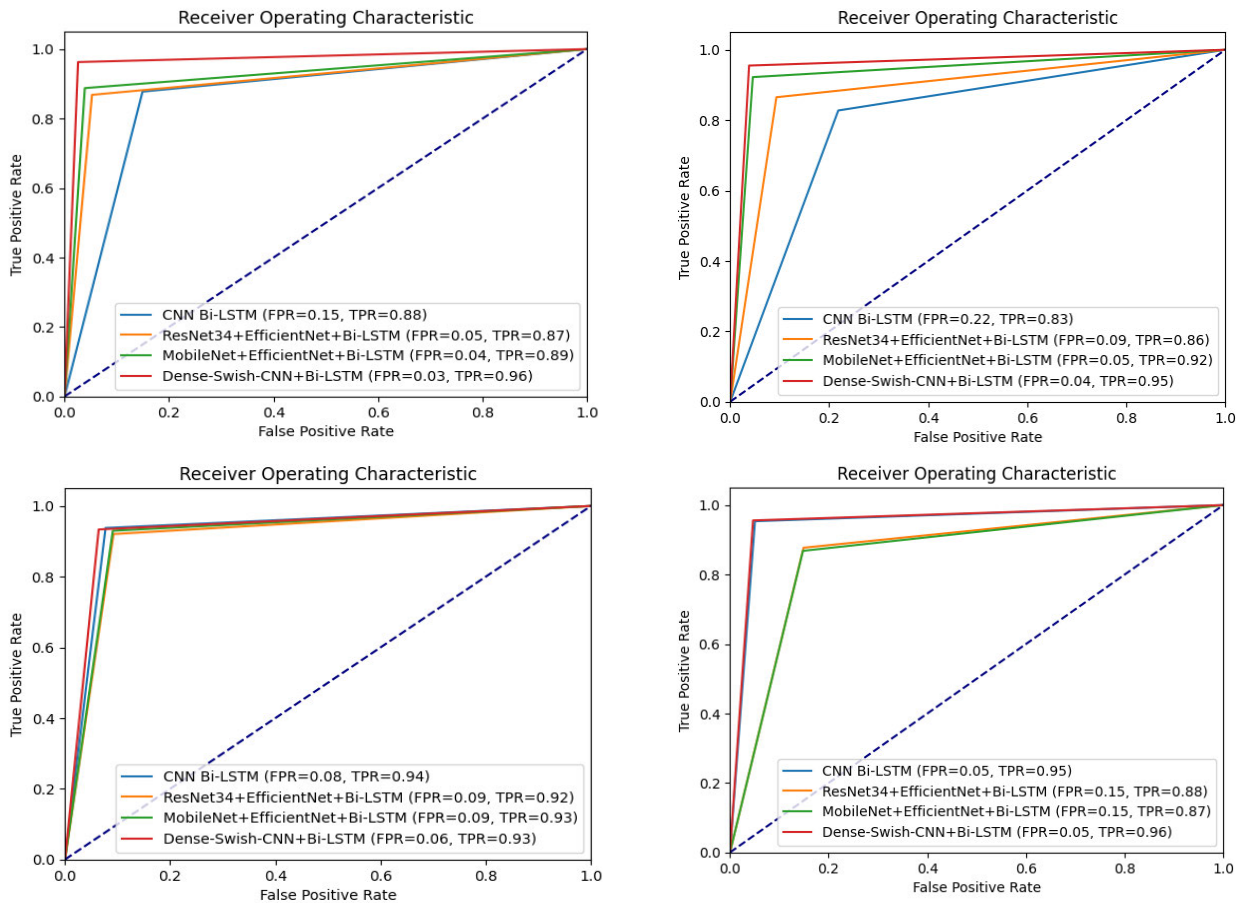


FIGURE 6. ROC curves for the proposed Dense-Swish-CNN over dataset.

obtained for all real and fake databases, with the outcomes illustrated in Figure 4. This evaluation provides insights into the effectiveness of our proposed approach in accurately distinguishing between real and fake images across different datasets.

Figure 5 shows the Deepfake detection performance, assessed in terms of accuracy for all CNN-based methods. According to the findings, the Dense-Swish-CNN-based Bi-LSTM strategy had the best accuracy out of all the models. On the other hand, the CNN Bi-LSTM method showed the least accuracy. It's worth noting that reducing the rates of false negatives is a primary objective of Deepfake recognition models, ensuring effective detection of manipulated content.

The detailed performance of each model is then assessed using the Bi-LSTM on the four datasets; the corresponding values for each performance metric are shown in Table 2 (DFDC), III (CelebDF), IV (Forgery NIR), and V(UADFV). We determined the recall rate for each pre-trained CNN model to evaluate the models. The Bi-LSTM method, which utilizes Dense-Swish-CNN as its foundation, exhibited the least false negative rates. Compared to the Convolutional Bi-LSTM method, it yielded the highest recall evaluation measure and the lowest recall value. Furthermore, reducing the false positive rate in Deepfake detection is crucial to

reducing the possibility of mistakenly classifying original content as Deepfakes. Consequently, we evaluated the accuracy of the proposed model for every CNN framework that was used.

Moreover, to thoroughly evaluate the Deepfake detection performance of the suggested method, we calculated the F1-score. Higher F1-scores indicate superior recognition ability. The Dense-Swish-CNN-based Bi-LSTM approach achieved the highest F1 score, while the CNN Bi-LSTM approach attained the lowest. Overall, all employed deep learning frameworks effectively recognized visual manipulations, highlighting the robustness of spatiotemporal frameworks in detecting alterations introduced in images to propagate false information.

The Receiver Operating Characteristic (ROC) curves for the Dense-Swish-CNN model assessed on the suggested datasets are shown in Figure 6. A ROC curve is a visual representation of the model's performance at various categorization thresholds. Contrasting the True Positive Rate (TPR) vs the False Positive Rate (FPR) helps see how different threshold settings affect the trade-offs between sensitivity and specificity.

Following, we discussed the classification results for all networks across all datasets to assess the recognition ability

TABLE 6. Performance of existing methods on generalization datasets.

Ref	Method	Dataset	Accuracy %
Matern et al. (2019) [30]	Logistic Regression MLP	UADFV	70.2
Ammar Elhassan et al. (2022) [35]	CNN		89.7
	DenseNet121		67.59
	DenseNet169		83.39
	EfficientNetB0		47.79
	ResNet50		46.29
	Vgg16		86.19
	Xception		
Zhiming Xia (2022) [36]	MesoNet		94.8
V. Premanand et al. (2024) [37]	SVM		82
Pranjal Ranjan, (2020) [38]	Transfer Learning Based CNN Framework	CelebDF	94.60
Umur Aybars et al. (2020) [23]	FakeCatcher		91.50
Aditi Kohli1 et al. (2020) [39]	frequency CNN		66.50
Pu et al. (2022) [40]	ResNet50		70.3
Yingxin et al. (2024) [41]	Generalized Multi-Scenario Deepfake Detection framework		83.16
Yukai et al. (2022) [42]	Xception	ForgeryNIR (MIX)	96.15
	GANFingerprint		94.76
	CNNDetection		97.56
	GANDCTAnalysis		64.82
	ML		29.00
Yukai Wang et al. (2023) [43]	CNNDetection		89.55
	FCAN-DCT + ResNet50		91.40
	FCAN-DCT + Xception		90.96
Ranjen et al. (2022) [38]	XceptionNet	DFDC	84.7
Ganguly et al. (2022) [44]	VixNet		95.42
Dessen et al. (2021) [45]	Convolutional vision transfer		91.5
Proposed	Dense-Swish-CNN+Bi-LSTM	DFDC CelebDF ForgeryNIR UADFV	97.76 96.98 96.5 96.91

of the proposed approaches.. Figure 7 presents the accuracy values obtained for all models, providing a clear performance overview on class-wise datasets. Notably, the Dense-Swish-CNN approach and the Bi-LSTM framework exhibited the highest class-wise results for categorizing real and fake data.

The efficacy of the suggested Dense-Swish-CNN-based Bi-LSTM technique for Deepfake detection across many genuine picture datasets is shown by the performance evaluation in Table 4. This is the most reliable method when considering model complexity and classification outcomes. One notable aspect of the Dense-Swish-CNN with Bi-LSTM classification method is its utilization of the swish activation function. This activation function allows the model to learn more visual features under diverse sample-capturing conditions. Consequently, the model gains a better understanding

of image alterations, which ultimately enhances the accuracy of Deepfake detection. By leveraging the strengths of Dense-Swish-CNN and Bi-LSTM, this approach achieves promising results in detecting DeepFake images within real image databases.

We use a binary classification setting and compare our proposed method with existing face manipulation detection algorithms on different datasets. The findings, which are shown in Table 6, show that our proposed framework performs better than current approaches in terms of accuracy across various datasets.

Compared to other recent approaches, the performance analysis conducted on four challenging datasets has convincingly demonstrated our approach's effectiveness in accurately recognizing manipulated visual samples.

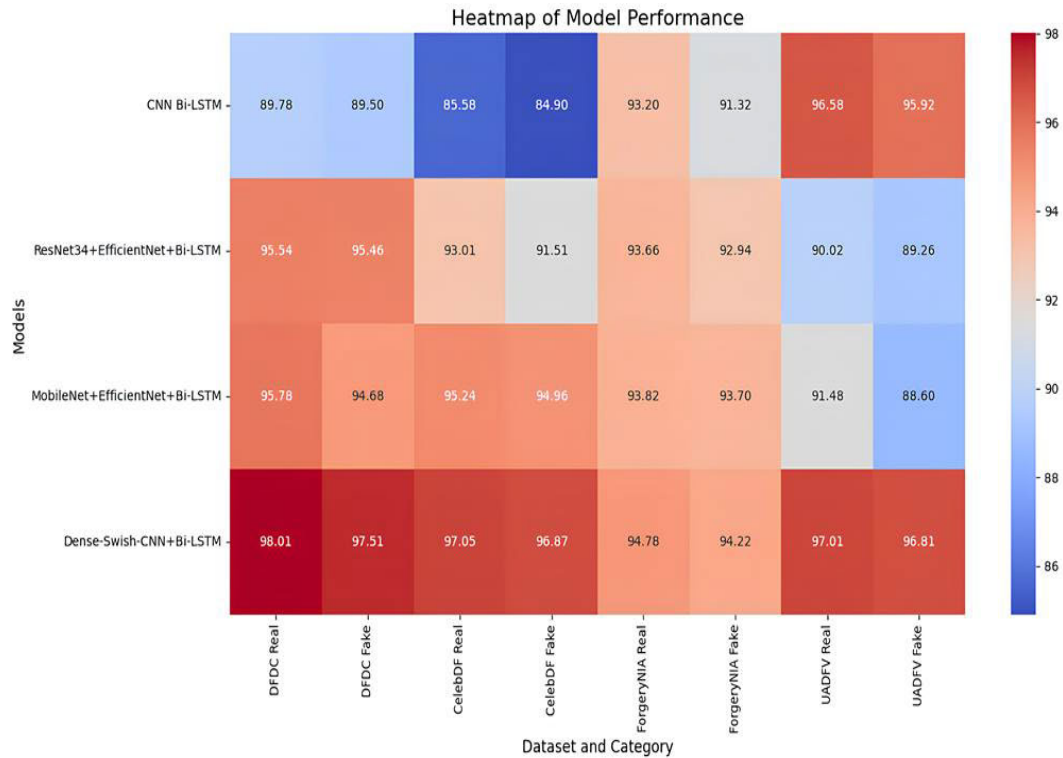


FIGURE 7. Heat map visualization of Class-wise comparative study.

Notably, our technique exhibits great performance on the DFDC dataset, suggesting its strong generalization ability. A vital advantage of the Dense-Swish-CNN approach is its ability to propagate negative scores during feature computation, resulting in the extraction of a denser and more discriminative set of visual characteristics. This enhancement contributes to the superior classification results achieved by our approach compared to other techniques.

VI. CONCLUSION

This study introduced a deep learning-based approach, Dense-Swish-CNN with Bi-LSTM, designed to detect real or deepfake samples from input data. Our method integrates spatial and temporal information of visual samples to identify alterations and distinguish between genuine and manipulated content effectively. We have utilized several base models, including CNN Bi-LSTM, ResNet34+EfficientNet+Bi-LSTM, and MobileNet+EfficientNet+Bi-LSTM, to compute frame-level information. Additionally, we proposed a novel deep learning-based feature extractor called Dense-Swish-CNN with Bi-LSTM. To evaluate the performance, we conducted experiments on four challenging datasets: DFDC, CelebDF, Forgery NIR, and UADFV. Our approach achieved the highest accuracy, with values of 97.76%, 96.98%, 96.5%, and 96.91% over the respective databases. Moving forward, we aim to extend our approach to other challenging video datasets and explore the effectiveness of

different models with the Bi-LSTM approach to enhance video deepfake detection performance further.

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